# Task 6 — Electronics & Hardware

## 1. Labelled Wiring Diagram & Architecture Description

The system utilizes a **split-rail power architecture** separating high-current inductive motor loads from sensitive digital logic and sensing circuits.

### Interconnection & Wiring Architecture

- **Power Source & Switching:** A Bonka 7.4 V 2S LiPo battery connects through an illuminated KCD3-3-101N SPST rocker switch to distribute switched 7.4 V power to the main power bus.
- **High-Current Motor Rail (7.4 V):** Switched battery power directly drives the motor supply pin (VM) of the TB6612FNG dual motor driver and feeds the input (IN) of the LM2596S step-down buck converter.
- **Intermediate Regulated Rail (5.0 V):** The LM2596S steps down the 7.4 V input to a stable 5.0 V DC output, which connects to the VIN pin of the ESP32-WROOM-32 development board.
- **Low-Voltage Logic Rail (3.3 V)**: The ESP32's onboard LDO regulator supplies a clean 3.3 V output (3V3 pin) to power the Pololu QTR-8A reflectance sensor array and the logic supply pin (VCC) of the TB6612FNG driver.
- **Common-Ground Topology:** A star-ground / common-ground hub links the battery negative terminal (BAT-), LM2596 ground, TB6612FNG ground pins (GND / PGND), ESP32 ground, and QTR-8A ground. This eliminates ground loops and prevents voltage offset errors in analog sensor readings.
- **Control & Sensing Interconnections:**
    - PWM & Direction: ESP32 GPIO pins connect to TB6612FNG control inputs (PWMA, AIN1, AIN2, PWMB, BIN1, BIN2) for independent speed and direction control of the two GA12-N20 gear motors.
    - Sensor Telemetry: Eight analog outputs from the QTR-8A array connect to the ESP32's ADC1 channels (GPIO 34, 35, 32, 33, 25, etc.) for real-time weighted centroid line position estimation.

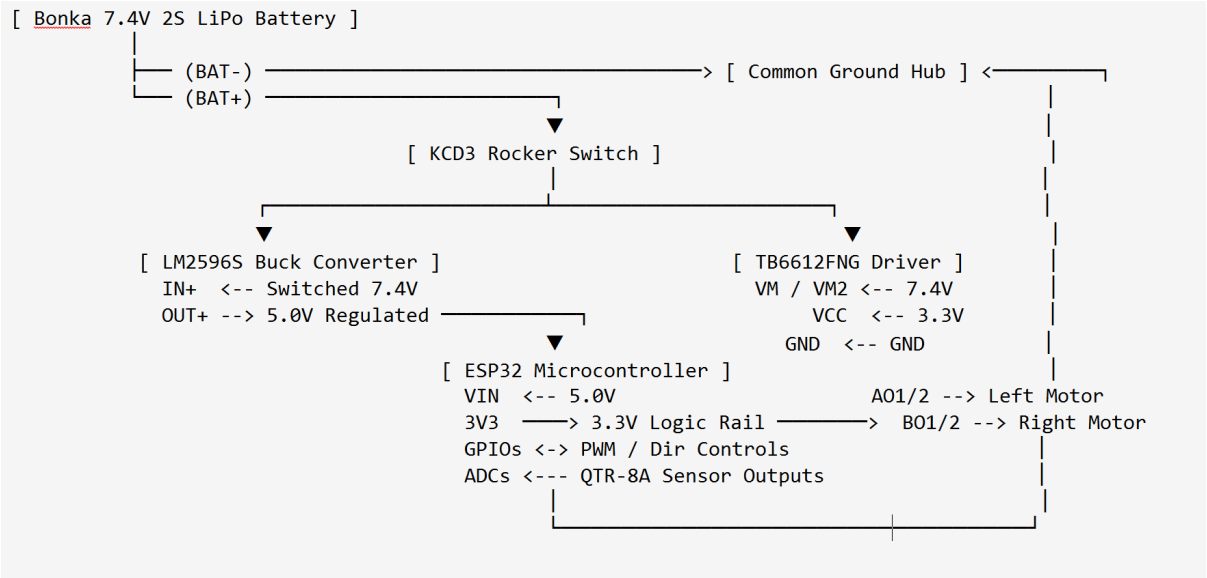
## 2. Functional Role of the Motor Driver (TB6612FNG)

The **TB6612FNG** is a dual-channel DC motor driver IC utilizing low-ON-resistance MOSFET output structures (LD MOS). Its primary functions in the robot include:

- **Current Amplification & Level Shifting:** Microcontroller GPIO pins (ESP32 at 3.3 V logic) cannot supply the current or voltage required to drive DC motors directly. The TB6612FNG bridges low-power control signals with the higher-voltage 7.4 V motor power rail.

- **H-Bridge Speed & Direction Control:** By accepting independent PWM signals and dual directional input logic (IN1/IN2), it enables forward rotation, reverse rotation, short-braking, and stop/standby modes for both left and right GA12-N20 motors.
- **Thermal & Undervoltage Protection:** It incorporates built-in thermal shutdown (TSD) and low-voltage detection (UVLO) circuitry to safeguard hardware against fault conditions.

## Wiring diagram




### Comprehensive System Wiring Table

| Subsystem / Module | Module Pin / Terminal | Connects To | Target System Rail / Bus | Notes / Function |
|---|---|---|---|---|
| **Bonka 7.4V 2S LiPo Battery** | BAT+ (Red Wire) | Rocker Switch Input (KCD3) | 7.4V Raw Bus | Main system power source. |
| | BAT- (Black Wire) | Common Ground Hub | System Ground (GND) | Common star-ground reference point. |

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| **KCD3 Rocker Switch** | Output Terminal | TB6612FNG VM/VM2 & LM2596 IN+ | Switched 7.4V Bus | Master ON/OFF power control. |
| **LM2596S Buck Converter** | IN+ / IN- | Switched 7.4V Bus & Ground Hub | 7.4V / GND | Steps down raw battery voltage to 5.0V. |
| | OUT+ | ESP32 VIN Pin | 5.0V Regulated Rail | Stable input supply for the microcontroller. |
| | OUT- | Common Ground Hub | System Ground (GND) | Buck converter ground return. |
| **ESP32-WROOM-32 MCU** | VIN | LM2596 OUT+ | 5.0V Regulated Rail | Powers the ESP32 board regulator. |
| | 3V3 | QTR-8A VCC & TB6612 VCC | 3.3V Logic Rail | Supplies clean logic/sensor power. |
| | GND | Common Ground Hub | System Ground (GND) | Digital ground reference. |
| | GPIO 14 | TB6612FNG PWMA | Control Signal | Left motor PWM speed control. |
| | GPIO 27 | TB6612FNG AIN1 | Control Signal | Left motor direction control A. |

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| | GPIO 26 | TB6612FNG AIN2 | Control Signal | Left motor direction control B. |
| | GPIO 13 | TB6612FNG PWMB | Control Signal | Right motor PWM speed control. |
| | GPIO 12 | TB6612FNG BIN1 | Control Signal | Right motor direction control A. |
| | GPIO 11 | TB6612FNG BIN2 | Control Signal | Right motor direction control B. |
| | GPIO 34 (ADC1_6) | QTR-8A Sensor Output 1 | Analog Signal | Leftmost line reflectance reading. |
| | GPIO 35 (ADC1_7) | QTR-8A Sensor Output 2 | Analog Signal | Line reflectance reading. |
| | GPIO 32 (ADC1_4) | QTR-8A Sensor Output 3 | Analog Signal | Center-left line reflectance reading. |
| | GPIO 33 (ADC1_5) | QTR-8A Sensor Output 4 | Analog Signal | Center-right line reflectance reading. |
| | GPIO 25 (ADC1_8) | QTR-8A Sensor Output 5 | Analog Signal | Rightmost line reflectance reading. |
| **TB6612FNG Motor Driver** | VM / VM2 | Switched 7.4V Bus | 7.4V Motor Rail | High-current power supply for H-bridge. |

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|---|---|---|---|---|
| | VCC | ESP32 3V3 Pin | 3.3V Logic Rail | Internal logic supply. |
| | GND / PGND1/2 | Common Ground Hub | System Ground (GND) | Power ground return for motor currents. |
| | STBY | 3.3V Rail (or ESP32 GPIO) | Logic HIGH | Pulls driver out of standby mode. |
| | AO1, AO2 | GA12-N20 Left Motor Terminals | Motor Power | Differential drive output for left wheel. |
| | BO1, BO2 | GA12-N20 Right Motor Terminals | Motor Power | Differential drive output for right wheel. |
| **Pololu QTR-8A Array** | VCC | ESP32 3V3 Pin | 3.3V Logic Rail | IR emitter and phototransistor power. |
| | GND | Common Ground Hub | System Ground (GND) | Sensor ground return. |
| | LEDON | ESP32 GPIO (or 3.3V) | Control Signal | Optional PWM brightness / power-saving control. |
| **GA12-N20 Motors (× 2)** | Motor Terminals (+/-) | TB6612 AO1/AO2 & BO1/BO2 | Power Out | Micro metal gear motors with encoders |

# 3. Troubleshooting Analysis: Controller Resetting When Motors Engage Symptom

The ESP32 microcontroller operates normally during stationary startup and sensor calibration, but **suddenly resets, reboots, or freezes** the moment high PWM duty cycles are applied to the motors (e.g., during rapid acceleration or sharp turn corrections).

## Probable Root Causes

1. **Input Voltage Sag / Brownout:** When motors abruptly draw high startup or stall current (up to $2.36A$ combined), the internal resistance of the LiPo battery or long, thin power wiring causes a transient voltage drop on the 7.4 V rail. This pulls the LM2596 input voltage below its minimum dropout threshold, causing the 5.0 V regulated output to dip below the ESP32's brownout detector threshold (~2.43 V).
2. **Inductive Voltage Spikes (Back-EMF):** Switching inductive DC motor windings generates high-frequency voltage transients that couple back into the power rails or ground plane, disrupting the logic supply if bypass capacitors are missing or improperly placed.
3. **Ground Bounce:** Inadequate copper trace width on shared ground returns causes a momentary voltage shift across the ground reference when motor currents surge, corrupting ADC readings and logic references.

## Diagnostic Checks & Confirmation

- **First Diagnostic Check:** Monitor the ESP32 3V3 or VIN power rail using a digital storage oscilloscope (DSO) or a multimeter with a fast min/max capture feature while engaging the motors. Alternatively, inspect the serial monitor for the ESP32 boot reason code (ESP_RST_BROWNOUT).
- **Confirmation of Root Cause:** If the oscilloscope reveals a voltage dip below $3.0V$ on the ESP32 VIN/3V3 pin concurrent with motor engagement, or if the serial monitor explicitly reports a Brownout detector was triggered, the root cause is confirmed as power rail voltage sag.

## Corrective Action & Fix Confirmation

1. **Bulk Capacitance:** Add a low-ESR electrolytic capacitor (470 uF to 680 uF, 25V) across the 7.4 V input rail near the LM2596 and TB6612FNG VM pins, along with a 10 uF capacitor on the 5.0 V regulated rail to absorb transient voltage drops.
2. **Decoupling:** Ensure 0.1 uF ceramic capacitors are soldered directly across the motor terminals to suppress brush sparking and high-frequency back-EMF spikes.
3. **Layout Optimization:** Widen all power and ground traces on the PCB or use star-ground wiring topology to eliminate parasitic inductive coupling.
4. **Fix Confirmation:** Re-run the motor engagement test at 100% PWM duty cycle. Confirm via serial telemetry that the ESP32 maintains stable operation without reboots, and verify on the oscilloscope that voltage sags remain well above the microcontroller's brownout threshold.